

BRONNOY, Norway

WMO: 011120

Lat: 65.457N

Lon: 12.218E

Elev: 25

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.0	(c) 20.8	(d) 4.1	(e) 6.8	(f) 21.7	(g) 7.3	(h) 8.0	(i) 23.1	(j) 39.0	(k) 33.1	(l) 35.0	(m) 32.4	(n) 7.5	(o) 40	(p) 0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	75.4	61.9	71.4	59.8	67.8	58.2	63.6	73.1	60.9	68.7	59.0	65.8	9.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	75.7	67.0	57.4	70.4	63.4	55.8	66.3	61.5	28.9	73.3	27.0	68.8	25.7	66.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 29.9	(b) 26.1	(c) 22.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 12.9	(b) 66.4	(c) 2.5	(e) 12.9	(f) 66.4	(g) 2.5	(h) 3.6	(i) 11.1	(j) 69.0	(k) 9.7	(l) 71.1	(m) 8.3	(n) 73.2	(o) 6.5	(p) 75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	32.5	33.5	34.8	40.0	47.0	51.6	57.0	56.7	52.7	44.9	38.3	35.0		
	DBStd	10.55	6.54	5.77	5.39	5.64	6.68	5.35	6.36	5.28	5.22	5.36	5.94	5.81		
	HDD50	2983	543	461	472	307	141	36	3	2	29	173	352	464		
	HDD65	7790	1008	881	937	750	558	402	267	265	369	623	801	929		
	CDD50	696	0	0	0	7	48	86	219	209	111	15	1	0		
	CDD65	30	0	0	0	0	1	2	19	7	1	0	0	0		
	CDH74	185	0	0	0	0	5	5	150	25	0	0	0	0		
	CDH80	27	0	0	0	0	0	0	27	0	0	0	0	0		
	WSAvg	10.3	12.0	12.4	11.7	10.0	9.4	9.0	7.7	8.3	9.1	10.7	11.5	12.4		
	PrecAvg	55.5	5.3	4.8	5.0	3.5	2.8	3.6	3.9	4.1	5.9	6.3	4.8	5.9		
	PrecMax	69.3	9.9	10.7	10.7	6.3	5.6	6.9	8.5	7.7	14.6	11.8	9.6	10.5		
(15) Precipitation	PrecMin	43.2	0.0	0.0	1.8	0.9	1.0	1.1	1.9	1.1	1.7	1.6	0.8	0.3		
	PrecStd	7.8	2.5	2.6	2.4	1.7	1.0	1.7	1.8	1.6	3.2	2.4	2.8	2.8		
	DB	46.7	46.6	48.4	63.4	72.4	75.0	83.2	77.2	69.5	61.3	54.3	47.5			
	MCWB	40.9	41.2	40.2	49.5	58.3	57.6	67.8	62.1	57.6	52.4	48.1	41.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	44.7	44.5	45.0	56.0	67.5	69.7	78.3	73.3	66.2	57.7	50.2	45.8			
	2% MCWB	41.8	41.3	40.5	45.6	54.7	57.1	63.4	60.6	57.0	50.2	45.7	43.0			
	5% DB	43.0	42.9	43.4	51.6	62.6	64.6	73.1	69.6	63.8	55.3	48.4	44.7			
	5% MCWB	40.3	40.0	40.4	43.8	52.7	55.2	62.5	59.3	55.3	49.1	44.3	42.1			
	10% DB	41.4	41.3	42.4	48.3	57.5	60.9	68.0	66.2	60.9	52.3	46.6	43.1			
	10% MCWB	38.5	38.4	39.7	41.7	49.7	53.3	59.8	57.8	54.2	47.4	42.8	40.2			
	0.4% WB	44.1	43.8	43.8	50.1	59.5	60.8	69.1	64.5	60.1	53.8	49.8	45.8			
	0.4% MCDB	44.6	44.6	44.3	62.2	71.7	70.4	80.6	74.3	66.7	59.5	51.3	46.0			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	42.4	42.1	42.4	46.8	55.5	57.9	65.5	61.6	57.8	51.7	47.0	44.4			
	2% MCDB	43.3	43.4	43.3	53.5	65.6	66.7	76.0	70.6	64.1	56.5	49.5	45.0			
	5% WB	40.8	40.6	41.2	44.6	53.0	55.9	62.8	59.8	56.4	50.0	45.1	42.4			
	5% MCDB	42.3	42.2	42.9	49.3	61.4	63.3	71.1	67.5	61.9	53.6	47.7	43.7			
	10% WB	38.7	38.6	39.5	42.9	50.4	54.0	60.2	58.3	54.8	48.4	42.9	40.2			
	10% MCDB	41.0	40.7	42.0	46.5	56.8	59.6	66.6	64.6	59.5	51.8	46.1	42.6			
	MDBR	6.7	6.9	7.9	9.5	11.0	9.9	11.1	10.5	9.9	7.9	6.5	6.5			
	5% DB	7.1	7.8	7.7	14.4	18.7	17.8	21.2	16.5	13.7	10.6	7.8	7.7			
(36) Mean Daily Temperature Range	MCDBR	6.3	7.0	6.4	8.1	9.0	8.9	9.5	8.3	7.1	6.6	5.9	7.2			
	5% WB	7.1	7.6	7.0	11.9	17.4	15.3	19.5	14.7	12.1	9.0	7.7	7.6			
	MCWBR	7.0	7.6	6.8	7.5	9.4	8.2	9.3	8.0	7.0	6.8	6.9	7.8			
	MCWBR	7.0	7.6	6.8	7.5	9.4	8.2	9.3	8.0	7.0	6.8	6.9	7.8			
(40) Clear-Sky Solar Irradiance	taub	0.367	0.296	0.324	0.348	0.348	0.347	0.355	0.342	0.330	0.305	0.346	0.364			
	taud	2.570	2.203	2.282	2.331	2.429	2.494	2.499	2.523	2.535	2.369	2.535	2.527			
	Ebn at Noon	80	182	236	259	272	275	269	262	239	182	77	23			
	Edh at Noon	7	17	25	30	30	30	29	25	20	15	6	3			
(44) All-Sky Solar Radiation	RadAvg	27	150	550	1090	1446	1537	1433	1120	598	242	50	7			
	RadStd	3	28	86	117	100	187	170	119	83	42	8	1			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page